

Algorithm Comparison

Compare GRN inference algorithm performance on hESC dataset

Select Algorithms to Compare

● GENIE3 ✓

● PPCOR ✓

● SINCERITIES ✓

● PIDC

● GRNBoost2

● SCENIC

Select All

Clear

Performance Metrics

Comparative performance across all metrics

↓ Export Data

Algorithm ↑↓	AUPRC ↑↓	AUROC ↑↓	F1 Score ↑↓	Early Precision ↑↓	Runtime (s) ↑↓
● GENIE3	0.847	0.923	0.782	0.891	252s
● GRNBoost2	0.803	0.891	0.756	0.867	324s
● SCENIC	0.789	0.878	0.741	0.852	486s
● SINCERITIES	0.734	0.856	0.698	0.801	522s
● PIDC	0.691	0.812	0.654	0.778	186s
● PPCOR	0.612	0.789	0.601	0.723	108s

Precision-Recall Curves

Multi-algorithm overlay

ROC Curves

Receiver operating characteristic

Top Motif Enrichment

Transcription factor binding site enrichment

Algorithm Similarity

Network prediction overlap (Jaccard index)

GENIE3 ↔ GRNBoost2

0.82

GENIE3 ↔ SCENIC

0.76

PPCOR ↔ PIDC

0.71

SINCERITIES ↔ SCENIC

0.68

Note:

High similarity indicates algorithms predict overlapping edge sets. Low similarity suggests complementary approaches.